LG-CISH: Lossless Coverless Image Steganography via CLIP
Semantic Hashing, Base-N Positional and Lehmer-Permutation Coding

Abstract

A steganographic scheme is proposed in which a secret message is transported by a sequence of ordinary, *unmodified* images rather than by perturbing the pixels of a single cover. Because no pixel is altered, the transmitted objects are statistically identical to natural images, and the classical trade-off between payload and detectability is, in effect, removed. The message is mapped to an ordered sequence of codebook images through two interchangeable, provably lossless channels: a base-N positional code that attains the information-theoretic $\log_2 N$ bits per image, and a Lehmer-permutation code that guarantees no image repeats within a block for a more natural-looking cover. Reliable recovery over lossy transport (re-compression, resizing, noise, cropping) is obtained by representing each codebook image with a normalised CLIP embedding and decoding by nearest neighbour, so that the correct index is recovered whenever a positive decoding margin is preserved. Confidentiality and integrity are added by a thin AES-256-CBC and CRC-32 layer, and cover plausibility is optionally improved by large-language-model-verified image aliases. This paper develops the information-theoretic motivation, the notation, and the full encoding and decoding procedures (Sections 1–3); the experimental evaluation is reported in later sections.

Keywords: coverless image steganography; information hiding; CLIP; semantic hashing; base-N coding; Lehmer permutation; steganalysis resistance.

1 Introduction

Steganography is the practice of hiding the very *existence* of a communication, in contrast to cryptography, which hides only its *content* [1]. The canonical setting is Simmons' prisoner's problem: two parties wish to coordinate an escape while every message they exchange is inspected by a warden, who releases a message only if it appears innocuous. A message is therefore safe not when it cannot be read, but when its carrier cannot be distinguished from ordinary traffic. This distinction motivates everything that follows, and it is made precise in this section using the language of probability and information theory. The notation used throughout the paper is collected in Table 1.

It is worth stating at the outset why the problem is treated distributionally rather than perceptually. A cover image that is altered only slightly may look identical to a human observer and yet be flagged with near certainty by a machine detector, because the detector reads statistics that the eye ignores. Conversely, an image that is transmitted without any alteration cannot be flagged by *any* detector on distributional grounds, however powerful, because it is literally a sample of the same population as innocent images. The method proposed in this paper is built entirely around this second observation, and the purpose of the present section is to explain, with the necessary mathematics, why the observation is decisive and what must be engineered so that it can be exploited in practice.

Table 1: Principal notation used throughout the paper.

Symbol	Meaning
$\mathcal{I}$	space of images (e.g. 512×512 RGB)
$P_C,\ P_S$	probability laws of cover and stego objects on $\mathcal{I}$
$m,\ \hat{m}$	secret message and its reconstruction
$k,\ \kappa$	steganographic / AES-256 key
$\mathcal{C} = \{c_0, \ldots, c_{N-1}\}$	shared codebook of N distinct images
$\Phi,\ e(\cdot)$	CLIP image encoder and its ℓ_2-normalised embedding
$s(a, b)$	cosine similarity of two embeddings
γ_t	per-image decoding margin (top-1 minus top-2 similarity)
$T(\cdot)$	lossy transport channel (JPEG, resize, noise, crop)
$L,\ B$	sequence length and permutation block size
$P(N, B)$	number of ordered B-image blocks, $N!/(N-B)!$
$D(\cdot\|\cdot)$	Kullback–Leibler divergence

1.1 The covert-communication model

Let a cover object be a realisation of a random image $X \in \mathcal{I}$ drawn from the natural-image law P_C. A classical (embedding) stegosystem is a triple of maps: an embedding function that produces a stego object,

$$Y = \mathrm{Emb}(X, m, k) \in \mathcal{I}, \tag{1}$$

an extraction function $\hat{m} = \mathrm{Ext}(Y, k)$, and a shared key k. The warden observes the transmitted object and applies a binary detector $f : \mathcal{I} \to \{\texttt{cover}, \texttt{stego}\}$. The security of the scheme is governed entirely by how far the law P_S of the transmitted object Y departs from the innocent law P_C. This warden channel is illustrated in Figure 1.

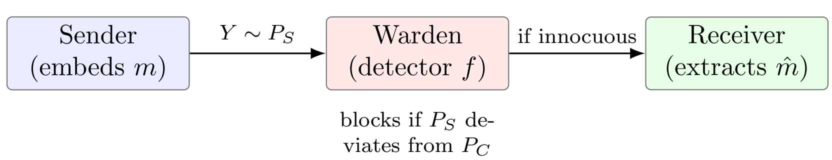

Figure 1: The prisoner's problem. The transmitted object passes only if the warden's detector cannot separate its law P_S from the natural-image law P_C. Security is a statement about distributions, not about pixels.

1.2 Threat model and assumptions

The setting adopted in this paper is made explicit through the following assumptions, which are standard for coverless steganography and are respected by the construction of Section 3.

Assumption 1.1 (Kerckhoffs' principle)**.** The warden is assumed to know the design of the stegosystem in full, including the coding, the feature map, and the cryptographic layer. Security is required to rest only on the secret quantities — the shared image database, the codebook ordering, and the AES key — and never on the obscurity of the algorithm.

Assumption 1.2 (Passive, computationally bounded warden)**.** The warden is passive: transmitted objects are inspected and either forwarded unchanged or blocked, but they are not maliciously rewritten. The warden may run any efficient detector, including learned steganalysers, but cannot break AES-256.

Assumption 1.3 (Shared secret database)**.** The sender and the receiver hold, offline, an identical and secret image database together with the same CLIP model. The database is treated as a shared key: an adversary without it cannot map images to indices. Loss of database secrecy reduces the scheme to its cryptographic layer only.

Assumption 1.4 (Benign but lossy channel)**.** The transport channel T degrades images through content-preserving operations — re-compression, resizing, mild noise, and moderate cropping — but does not deliberately destroy them. The degradation is bounded so that semantic content is retained, which is the regime in which Eqn (18) can be guaranteed.

Under Assumptions 1.1–1.4 the two objectives of the scheme can be separated cleanly: *undetectability* is an assertion about the marginal law of each transmitted image (§1.4–1.6), whereas *reliability* is an assertion about the recovery of indices from a degraded channel (§1.8). These are treated independently in the analysis that follows.

1.3 Embedding distortion and the payload it buys

Any embedding of the form of Eqn (1) changes at least one pixel, so a distortion is incurred between the cover X and the stego Y. For n-pixel images the mean squared error and the peak signal-to-noise ratio are

$$\mathrm{MSE}(X,Y) = \frac{1}{n}\sum_{i=1}^{n}\big(x_i - y_i\big)^2, \tag{2}$$

$$\mathrm{PSNR}(X,Y) = 10\,\log_{10}\frac{V_{\max}^2}{\mathrm{MSE}(X,Y)} \quad [\mathrm{dB}], \tag{3}$$

with $V_{\max} = 255$ for 8-bit images. Perceptual quality is frequently summarised by the structural similarity index

$$\mathrm{SSIM}(X,Y) = \frac{(2\mu_X\mu_Y + C_1)(2\sigma_{XY} + C_2)}{(\mu_X^2 + \mu_Y^2 + C_1)(\sigma_X^2 + \sigma_Y^2 + C_2)}, \tag{4}$$

where μ, σ denote local means, variances, and covariance, and C_1, C_2 are stabilising constants. The amount of secret data carried is measured by the relative payload (embedding rate)

$$\alpha = \frac{|m|}{n} \quad [\text{bits per pixel}]. \tag{5}$$

It is intuitively clear, and made precise below, that larger α forces larger distortion and hence a larger statistical footprint. This tension is the central obstacle that the present work seeks to avoid rather than merely to balance.

The trade-off is formalised by rate–distortion theory. For a given cover source there exists a capacity–distortion function $C(D)$, the largest secure payload achievable within a distortion budget D, which is non-decreasing and concave; a family of embedding methods differs only in how closely it approaches this frontier. Classical spatial and transform-domain schemes — least-significant-bit substitution, and DCT- or DWT-based embedding — all operate strictly inside the region $D > 0$, and thus purchase every additional bit with additional distortion and, through Eqn (9), with additional detectability. The proposal of this paper is to leave that frontier altogether by operating at the single point $D = 0$, which the coverless principle of §1.6 shows to be attainable.

1.4 Statistical detectability and its fundamental limits

The proper measure of steganographic security is not distortion but statistical distinguishability. Following Cachin, the (in)security of a stegosystem is quantified by the Kullback–Leibler divergence between the innocent and the transmitted laws,

$$D(P_C \,\|\, P_S) = \sum_{z \in \mathcal{I}} P_C(z) \, \log \frac{P_C(z)}{P_S(z)}, \tag{6}$$

and a scheme is called ε-secure when

$$D(P_C \,\|\, P_S) \le \varepsilon, \qquad \text{with perfect security at } \varepsilon = 0. \tag{7}$$

The operational meaning of Eqn (6) is exposed by hypothesis testing. For any warden detector f with false-alarm probability P_{FA} and missed-detection probability P_{MD}, the average error under equal priors is

$$P_E = \tfrac{1}{2}\big(P_{\mathrm{FA}} + P_{\mathrm{MD}}\big), \tag{8}$$

and this error is bounded below, through Pinsker's inequality, by the divergence in Eqn (6):

$$P_E \;\ge\; \frac{1}{2} - \sqrt{\tfrac{1}{2}\, D(P_C \| P_S)}. \tag{9}$$

Two consequences are read directly from Eqn (9). First, when $D(P_C \| P_S) \to 0$ the best possible detector is driven to $P_E \to \frac{1}{2}$, i.e. to blind guessing. Second, any embedding that perturbs the cover makes $P_S \neq P_C$ and therefore makes $D > 0$, which in turn caps how undetectable the scheme can be. A further limit, the square-root law of steganographic capacity, states that the largest payload that keeps D bounded grows only as the square root of the cover size,

$$|m^\star| = \Theta\big(\sqrt{n}\big)\,, \tag{10}$$

so that the *secure* embedding rate $\alpha^\star = |m^\star|/n = \Theta(1/\sqrt{n})$ actually vanishes as images grow. Modern convolutional steganalysers such as XuNet [4] and SRNet [5] approach these theoretical limits in practice, detecting embedding at rates once believed safe. Embedding steganography is thus squeezed from both sides: distortion is unavoidable (Eqn (3)), and even small distortion is increasingly detectable (Eqn (9)–10).

1.5 Detector performance and the cost of a single error

Two further quantities are used later and are introduced here for completeness. The discriminating power of a warden's soft detector, which outputs a score whose mean shifts by $\Delta\mu$ between cover and stego while retaining variance σ^2, is often summarised by the deflection coefficient

$$d^2 = \frac{(\Delta\mu)^2}{\sigma^2}, \tag{11}$$

a larger d^2 meaning an easier detection problem; for an unmodified transmitted object $\Delta\mu = 0$ and hence $d^2 = 0$. The overall quality of a detector across all thresholds is reported by the area under its receiver-operating-characteristic curve, AUC $\in [\frac{1}{2}, 1]$, with AUC $= \frac{1}{2}$ corresponding to the chance line of Eqn (15).

On the receiver's side, a complementary bound explains why losslessness is demanded rather than merely desired. Regarding the recovered index sequence as the output of a noisy channel whose input is the message, Fano's inequality relates the probability of a decoding error P_e to the residual uncertainty about the message,

$$H(m \mid \hat{m}) \;\le\; H_b(P_e) + P_e \,\log_2\!\big(|\mathcal{M}| - 1\big), \tag{12}$$

where $H_b(\cdot)$ is the binary entropy function and $\mathcal{M}$ the message alphabet. Moreover, because decoding is a deterministic function of the received images, the data-processing inequality guarantees that no post-processing can increase the information already present,

$$I(m; \hat{m}) \;\le\; I(m; \{\tilde{y}_t\})\,. \tag{13}$$

Equations (12)–(13) show that a single mis-recovered index injects irreducible uncertainty that later stages cannot repair; this is precisely why the design in Section 3 aims for $P_e = 0$ on a clean channel and, where the channel is too harsh, converts an error into a *detected* failure by the integrity check rather than allowing silent corruption.

1.6 The coverless principle: removing the footprint

The escape from this squeeze is to *not embed at all*. In coverless steganography no cover is modified; instead, an ordinary image is *selected* (or generated) so that some intrinsic property of the image already encodes the desired bits [3, 6]. The transmitted object is then a genuine sample of the natural-image law, which is the key distributional fact:

$$Y \in \operatorname{supp}(P_C) \text{ unmodified} \implies P_S = P_C \implies D(P_C \| P_S) = 0. \tag{14}$$

Substituting Eqn (14) into Eqn (9) yields the central promise of the paradigm,

$$P_E \;\ge\; \tfrac{1}{2}, \tag{15}$$

that is, undetectability holds *by construction* against any pixel-domain detector, because the transmitted images are, by definition, drawn from the same distribution as innocent images. The design problem is thereby transformed: the question is no longer "how may pixels be perturbed with least distortion?" but "how may a secret be *mapped* to a sequence of unmodified images, and *recovered* after that sequence has traversed a lossy channel?". The three paradigms are contrasted in Table 2.

Within the coverless family two branches are distinguished. In the *generative* branch the transmitted image is synthesised from the secret by a generative model, so that new pixels are produced; although no cover is *edited*, the synthetic image may still deviate from the natural-image law and the mapping is not always exactly invertible. In the *mapping* (selection) branch, which is adopted here, the transmitted image is an existing, unmodified database image chosen so that its index encodes the bits; the transmitted object is then a genuine natural image and Eqn (14) holds exactly. The mapping branch has historically relied on robust perceptual hashing to index the database, which is discriminative but fragile under geometric transport; the substitution of a semantic hash for a perceptual one, developed in §1.9, is the change that makes the mapping branch reliable over real channels.

Table 2: Comparison of steganographic paradigms. The coverless mapping paradigm, adopted here, is the only one that is both distortion-free and lossless.

Property	Embedding (LSB/DCT)	Coverless (generative)	Coverless (mapping, ours)
Modifies pixels	Yes	synthesises new pixels	**No**
$D(P_C \| P_S)$	> 0	small–moderate	0
Distortion (PSNR)	finite	finite	∞
Lossless recovery	usually	often not	**Yes**
Raw capacity	high	moderate	database-bound

1.7 The mapping formulation

Let a shared *codebook* $\mathcal{C} = \{c_0, c_1, \ldots, c_{N-1}\}$ of N mutually distinct images be fixed offline. A message is regarded as an integer and represented in a positional number system whose "digits" select codebook images. If the codebook is treated as a radix-N alphabet, then each transmitted image carries

$$C_{\mathrm{img}} = \log_2 N \quad \text{[bits per image]}, \tag{16}$$

so that a message requiring L images conveys $L \log_2 N$ bits in total. For the codebook used in this work, $N = 40$ gives $C_{\mathrm{img}} = \log_2 40 \approx 5.322$ bits per image. The mapping is a bijection between integers and index sequences, which is what makes the scheme *lossless*: no information is discarded at any stage of the encoding.

Two properties of this construction deserve emphasis. First, the assignment of indices to images is itself part of the shared secret: an adversary who observes the sequence but does not hold the codebook cannot invert the map, so the codebook acts as a large key and contributes the factor $N!$ to the keyspace of Eqn (28). Second, the map is *positional*, so that the same image placed in two different positions of the sequence contributes different amounts to the encoded integer, exactly as a digit contributes differently according to its place. It is therefore the *ordered* sequence, and not the multiset of images, that carries the message; the order is not incidental packaging but the substance of the code.

1.8 The channel and the recovery condition

Real messaging and social platforms do not deliver images untouched; they re-compress, resize, and re-encode them. A photograph forwarded through a popular messaging application, for example, is routinely re-encoded at a lower JPEG quality and downscaled to a bounded resolution, while the same image posted to a social network may be cropped to a fixed aspect ratio and stripped of its metadata. Any scheme that hides information in fragile low-level statistics is destroyed by such ordinary handling, which is why robustness to these operations is treated here as a first-class requirement rather than as an afterthought. This handling is modelled by a transport operator $T : \mathcal{I} \to \mathcal{I}$, so that the receiver observes distorted images $\tilde{y}_t = T(y_t)$. The receiver must map each distorted image back to the index the sender intended. Writing $c_{j^\star}$ for the image actually sent at position t, the decoded index is chosen as the codebook entry most similar to the received image,

$$\hat{\imath}_t = \arg\max_{0 \le j < N} \; s(e(\tilde{y}_t),\, e(c_j))\,, \tag{17}$$

where $e(\cdot)$ is a feature map (developed in §1.9) and $s(\cdot,\cdot)$ its similarity. Recovery at position t is correct precisely when the true image outscores every rival, which is captured by a positive *decoding* *margin*

$$\gamma_t = s(e(\tilde{y}_t), e(c_{j^\star})) - \max_{j \neq j^\star} s(e(\tilde{y}_t), e(c_j)) > 0. \tag{18}$$

If the feature map is stable under transport, in the sense that $\|e(T(x)) - e(x)\|_2 \leq \delta$ for all codebook images, and if the codebook is separated so that distinct images differ by at least 2δ in feature space, then Eqn (18) is guaranteed to hold and every index is recovered exactly. The reliability of the whole scheme therefore rests on choosing a feature map that is *robust to transport* yet *discriminative between images*.

The consequence of Eqn (18) for a whole message is multiplicative. If correct recovery at the L positions is treated as independent with per-image failure probability $p_t = \Pr[\gamma_t \leq 0]$, then the probability that the entire message is recovered without a single error is

$$\Pr[\text{message correct}] = \prod_{t=1}^{L} (1 - p_t) \;\approx\; \exp\left(-\sum_{t=1}^{L} p_t \right), \tag{19}$$

so that even a small per-image error rate is amplified by a long sequence. Two design responses follow, and both are adopted in Section 3. First, the feature map is chosen so that p_t is essentially zero across the operating range of the channel, which drives Eqn (19) towards one. Second, for degradations beyond that range the integrity check of Eqn (27) converts the residual failures into *detected* errors, so that a corrupted message is discarded and retransmitted rather than silently accepted.

1.9 Semantic hashing with a vision–language model

Perceptual hashes, built from low-level pixel or frequency statistics, satisfy discriminability but degrade quickly under geometric transport such as cropping. In this work the feature map is instead taken from CLIP, a vision–language model trained by contrastive learning on a very large corpus of image–text pairs [2]. Only its image encoder $\Phi : \mathcal{I} \to \mathbb{R}^d$ ($d = 512$) is used, and each image is represented by the ℓ_2-normalised embedding

$$e(x) = \frac{\Phi(x)}{\|\Phi(x)\|_2}, \qquad \|e(x)\|_2 = 1, \tag{20}$$

so that the similarity in Eqn (17) is the cosine (inner) product

$$s(a, b) = \langle a, b \rangle = a^\top b \in [-1, 1]. \tag{21}$$

Because CLIP is trained to place *semantically* similar images near one another, its embedding of a JPEG-compressed, resized, or partially cropped image stays close to the embedding of the original: the visual meaning survives even when the pixels do not. This property is exactly the transport stability required by §1.8, and it is the reason a semantic hash succeeds where a perceptual hash fails.

The robustness just described is not accidental but a direct consequence of the training objective. The encoder is trained to maximise the agreement between an image and its matching caption relative to mismatched captions, through a symmetric contrastive (InfoNCE) loss. For a batch of K image–text pairs with image embeddings v_i and text embeddings w_j and temperature ϑ, the image-to-text term of that loss is

$$\mathcal{L} = -\frac{1}{K} \sum_{i=1}^{K} \log \frac{\exp(\langle v_i, w_i \rangle / \vartheta)}{\sum_{j=1}^{K} \exp(\langle v_i, w_j \rangle / \vartheta)}, \tag{22}$$

with an analogous text-to-image term. Minimising Eqn (22) over a very large and varied corpus forces the image encoder to place all views of the same underlying scene — including compressed, rescaled, or lightly cropped views — close together, because such views share the same caption. The embedding is therefore approximately invariant to precisely the content-preserving degradations that make up the transport channel of Assumption 1.4, which is the mechanism that keeps the decoding margin of Eqn (18) positive in practice.

1.10 Capacity of the image-sequence channel

Two lossless codes are developed in this paper. The first, base-N positional coding, attains the per-image ceiling of Eqn (16) but permits an image to repeat within a sequence. A repeated image can make a cover look unnatural, so a second code is introduced in which every block of B images consists of *distinct* images chosen without repetition. The number of such ordered blocks is the falling factorial

$$P(N,B) = \frac{N!}{(N-B)!} = \prod_{t=0}^{B-1}(N-t), \tag{23}$$

and a block of B images therefore carries $\log_2 P(N,B)$ bits, i.e.

$$C_{\text{perm}} = \frac{1}{B}\log_2 P(N,B) \quad \text{[bits per image]}. \tag{24}$$

Using Stirling's approximation, $\log_2 N! \approx N\log_2(N/e)$, the full-block case $B = N$ satisfies $C_{\text{perm}} \approx \log_2(N/e)$, which is below the base-N ceiling of Eqn (16) by roughly $\log_2 e \approx 1.443$ bits per image; smaller blocks recover most of the loss. Thus Eqn (16) and Eqn (24) bound a controllable trade-off between raw capacity and cover plausibility, which is one of the design levers of the method.

1.11 The lossless requirement and integrity

Since a single mis-recovered index corrupts every subsequent bit, the scheme is designed so that, on a clean channel, reconstruction is bit-exact. Denoting the original and recovered payload bits by p_b and $\hat{p}_b$ over B_{tot} bits, the bit error rate is

$$\text{BER} = \frac{1}{B_{\text{tot}}}\sum_{b=1}^{B_{\text{tot}}} \mathbb{1}[\hat{p}_b \neq p_b], \tag{25}$$

and, because every coding stage is a bijection, BER $= 0$ whenever all indices are recovered ($\gamma_t > 0\ \forall t$, Eqn (18)). For the lossless case the mutual information between the sent and the reconstructed message attains the message entropy,

$$I(m;\hat{m}) = H(m), \tag{26}$$

i.e. the covert channel is noiseless end-to-end. To detect the rare event of a transport failure, a 32-bit cyclic redundancy check is appended, which flags a corrupted message with probability

$$\Pr[\text{undetected error}] \approx 2^{-32}. \tag{27}$$

Finally, confidentiality and key strength are provided by AES-256-CBC, giving a combined secret space of

$$K = N! \cdot 2^{256}, \tag{28}$$

the product of the number of codebook orderings and the AES key space.

1.12 Source coding, expected length, and a worked example

The number of images that a message occupies is governed by information theory. If the message is modelled as a source with entropy $H(m)$ bits, then, by Shannon's source-coding theorem, no lossless compressor can represent it in fewer than $H(m)$ bits on average,

$$\mathbb{E}[\,|\textsc{zlib}(m)|\,] \;\geq\; \frac{H(m)}{8} \quad \text{[bytes]}, \tag{29}$$

and the compressor of Eqn (32) is used precisely to push the encoded size towards this floor before any image is chosen. Once the framed payload occupies $|f|$ bytes, the base-N code of Eqn (16) produces an expected sequence length

$$\mathbb{E}[L] \;\approx\; \frac{8\,|f|}{\log_2 N}, \tag{30}$$

so that capacity and length are reciprocal: a larger codebook shortens the cover. A concrete illustration is given in Table 3 for the codebook used in this work ($N = 40$, $\log_2 N \approx 5.322$ bits per image). It is emphasised that the figures in Table 3 are dominated, for short messages, by the fixed overhead of the cryptographic frame (the CRC, the AES initialisation vector and padding, and the compression header); the per-image rate of 5.322 bits is the asymptotic behaviour that governs long messages.

Table 3: Worked example of Eqn (30) for the $N = 40$ codebook. The framed size includes compression, a 4-byte CRC, and AES padding; short messages are overhead-dominated, whereas long messages approach $8|f|/\log_2 40$ images.

Message class	Payload bits	Images $\approx 8\|f\|/\log_2 40$	Rate regime
Short ($\leq$50 chars)	$\sim$300	$\sim$57	overhead-dominated
Medium (50–200 chars)	$\sim$800	$\sim$150	transitional
Long (200–1000 chars)	$\sim$2100	$\sim$400	asymptotic (5.322 b/img)

1.13 Problem statement, design goals, and contributions

The preceding development frames the problem addressed in this paper. *Given* a shared image database and a shared vision–language model, a scheme is sought that (i) transmits only unmodified natural images, so that Eqn (14)–15 hold and detection is reduced to chance; (ii) is lossless, so that Eqn (25)–26 hold on a clean channel; and (iii) recovers correctly after realistic transport, so that Eqn (18) is satisfied across compression, noise, resizing, and cropping. The following contributions are made:

1. A coverless mapping scheme, **LG-CISH**, is proposed in which the identity and order of unmodified images carry the secret, and reliable recovery is obtained by CLIP semantic hashing (§1.9, §3).

2. Two interchangeable lossless codes are formalised — base-N positional coding at the $\log_2 N$ ceiling (Eqn (16)) and Lehmer-permutation coding with a no-repeat guarantee (Eqn (23)–24) — together with a whitening step that removes an ordering artefact.

3. A confidentiality and integrity layer (AES-256-CBC, CRC-32) and an optional large-language-model-verified alias mechanism for cover plausibility are integrated without affecting losslessness (§3.1.2, §3.3).

1.14 Organisation of the paper

The remainder of the paper is organised as follows. Related work is reviewed in Section 2. The proposed method — notation and shared state, cryptographic framing, the two coding channels, and the encoding and decoding procedures — is developed in Section 3. The experimental setup, results, robustness, steganalysis resistance, ablation, comparison, and statistical validation are reported in the subsequent sections.

2 Literature Review

[This section is intentionally left blank in the present draft and will be written subsequently.]

3 Proposed Method

3.1 Overview

The proposed method is a lossless, coverless pipeline in which a secret message is carried by an ordered sequence of unmodified codebook images. The sender and the receiver are assumed to share, offline, three artefacts: an image database, the CLIP image encoder, and the resulting codebook that assigns a unique index to each database image. On the sender side the message is first compressed, integrity tagged, and optionally encrypted, then converted into a sequence of codebook indices by one of two lossless codes, and finally emitted as the corresponding images. The images traverse an arbitrary lossy channel. On the receiver side each received image is embedded with CLIP and matched to the nearest codebook entry to recover its index, after which the coding and cryptographic steps are inverted to reconstruct the message exactly. The end-to-end flow is shown in Figure 2. Each component is developed in turn in the following subsections.

The architecture is organised around a deliberate separation of concerns, each concern being handled by a distinct stage so that the stages can be reasoned about and modified independently. *Confidentiality and integrity* are handled first, by the cryptographic frame of §3.1.2, so that everything downstream operates on an opaque, integrity-protected byte string and the coding layer need know nothing about the message semantics. *Capacity and cover plausibility* are handled next, by the choice between the two coding channels of §3.2, which converts the byte string into indices at a controllable rate and with an optional no-repeat guarantee. *Undetectability* is inherited for free from the coverless principle, because the emitted objects are unmodified database images. Finally, *reliability* is delegated entirely to the semantic feature map at decode time, so that the sender performs no neural computation at all and the burden of channel robustness falls on a single, well-understood component.

The ordering of the pipeline is not arbitrary. Compression is applied before encryption because encrypted data is effectively incompressible, so any compression must precede the cipher; the integrity tag is computed over the compressed payload so that a corruption anywhere in transit is caught before decompression is attempted; and the positional coding is applied last, on the fully framed bytes, so that it is agnostic to whether encryption was used. This layered discipline is what allows the losslessness argument of Proposition 3.1 to be stated as a simple composition of invertible stages.

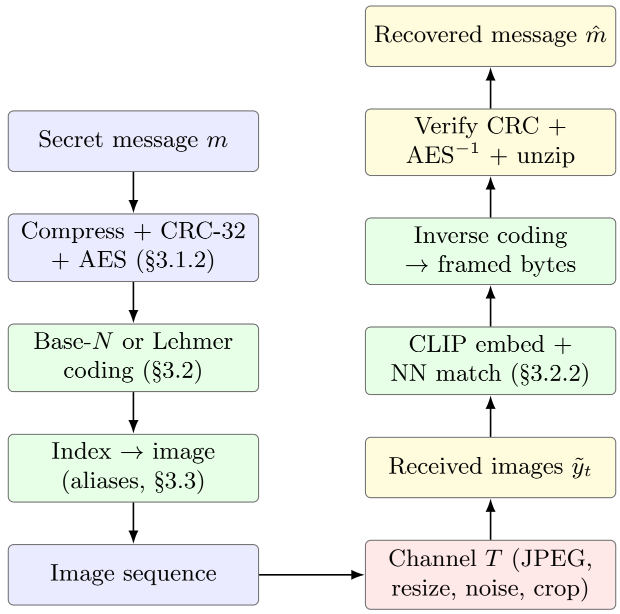

Figure 2: End-to-end flowchart of LG-CISH. The left column is the sender, the right column the receiver; the two are joined by the lossy channel T. Every transformation is deterministic and invertible, so the covert channel is lossless whenever the indices are recovered correctly.

3.1.1 Pre-processing: notation and shared state

The shared codebook is denoted $\mathcal{C} = \{c_0, \ldots, c_{N-1}\}$, and its CLIP embeddings are stored as the rows of a matrix $\mathbf{C} \in \mathbb{R}^{N \times d}$ with $\mathbf{C}_j = e(c_j)$, where $e(\cdot)$ is the normalised embedding of Eqn (20). So that the nearest-neighbour rule of Eqn (17) is unambiguous, the codebook is required to be *separated*: no two distinct images may exceed a similarity threshold τ,

$$\max_{i \neq j} \; s(\mathbf{C}_i, \mathbf{C}_j) \leq \tau, \qquad (\tau = 0.85). \tag{31}$$

A candidate database is reduced to a separated codebook by the greedy construction of Algorithm 1, which retains an image only if it is not too similar to any image already kept.

The size N of the codebook is a design parameter that trades capacity against the difficulty of maintaining separation. By Eqn (16) the per-image capacity grows only logarithmically in N, so doubling the codebook adds just one bit per image; yet as N grows it becomes progressively harder to keep every pair of images below the threshold τ of Eqn (31), because the embedding space becomes more densely populated and the minimum pairwise separation shrinks. A moderate codebook is therefore preferred: it keeps the separation margin $1 - \max_{i \neq j} s(\mathbf{C}_i, \mathbf{C}_j)$ comfortably positive, which is exactly the quantity that, together with the transport stability of Assumption 1.4, guarantees correct recovery through Definition 3.1. The threshold τ is thus best read not as an arbitrary constant but as the knob that sets how much channel degradation the codebook is provisioned to absorb.

Definition 3.1 (Separated codebook)**.** A codebook $\mathcal{C}$ is τ-separated if every pair of its embeddings satisfies Eqn (31). Under τ-separation and transport stability $\|e(T(c_j)) - e(c_j)\|_2 \leq \delta$ with $2\delta < 1 - \tau$, the recovery rule of Eqn (17) returns the correct index for every codebook image.

Algorithm 1 Codebook construction (pre-processing)

Input: candidate image set $\mathcal{D}$, encoder Φ, threshold τ
Output: codebook embeddings $\mathbf{C}$, image paths, size N
$E \leftarrow [\, e(x) \text{ for } x \in \mathcal{D}\,]$ ▷ *normalise, Eqn* (20)
$\mathcal{K} \leftarrow \emptyset$
for $j = 0$ **to** $|\mathcal{D}| - 1$ **do**
    if $\max_{i \in \mathcal{K}} s(E_i, E_j) \le \tau$ **then** $\mathcal{K} \leftarrow \mathcal{K} \cup \{j\}$ ▷ *Eqn* (31)
end for
return $\mathbf{C} \leftarrow E[\mathcal{K}]$, paths$[\mathcal{K}]$, $N \leftarrow |\mathcal{K}|$

3.1.2 Cryptographic framing

Before any image is chosen, the message is wrapped in a thin layer that provides compression, integrity, and confidentiality. The message is first losslessly compressed with zlib, which shortens the sequence; a 32-bit CRC is then appended for integrity; and the result is optionally encrypted with AES-256-CBC under a 256-bit key κ, with a random 16-byte initialisation vector prepended to the ciphertext. Writing these operations explicitly, the framed payload is

$$p_0 = \textsc{zlib}(\textsc{utf8}(m))\,, \tag{32}$$

$$f = \textsc{aes}_\kappa(p_0 \,\|\, \textsc{crc}32(p_0))\,, \tag{33}$$

where the AES step reduces to the identity when no key is supplied, in which case integrity is still provided by the CRC. The encryption itself follows the cipher-block-chaining recurrence: with the padded, framed plaintext split into 16-byte blocks $t_1, t_2, \ldots$, a random initialisation vector $c_0 = IV$, and the block cipher E_κ, the ciphertext blocks are produced by

$$c_i = E_\kappa(t_i \oplus c_{i-1})\,, \qquad i = 1, 2, \ldots, \tag{34}$$

and are recovered by the inverse recurrence $t_i = E_\kappa^{-1}(c_i) \oplus c_{i-1}$. The chaining in Eqn (34) ensures that two identical plaintext blocks do not, in general, encrypt to identical ciphertext blocks, so that no block-level regularity of the payload survives into the emitted index stream. The inverse operation, unwrap, decrypts, separates the trailing four CRC bytes, and *rejects* the message if the recomputed check disagrees, thereby realising the guarantee of Eqn (27). Cipher-block chaining is used so that identical plaintext blocks do not produce identical ciphertext blocks; the encryption is summarised in Algorithm 2.

Algorithm 2 Cryptographic framing (wrap / unwrap, AES-256-CBC)

function $\textsc{wrap}(m, \kappa)$
    $p_0 \leftarrow \textsc{zlib}(\textsc{utf8}(m))$ ▷ *Eqn* (32)
    $t \leftarrow p_0 \,\|\, \textsc{crc}32(p_0)$ ▷ *append 4-byte CRC*
    if κ given **then** $IV \leftarrow \textsc{rand}(16)$; **return** $IV \,\|\, \textsc{aes-cbc}_\kappa(\textsc{pad}(t), IV)$
    else return t ▷ *CRC-only mode*
function $\textsc{unwrap}(f, \kappa)$
    if κ given **then** $t \leftarrow \textsc{unpad}(\textsc{aes-cbc}_\kappa^{-1}(f_{[16:]}, f_{[:16]}))$ **else** $t \leftarrow f$
    split $t = p_0 \,\|\, c$; **if** $c \ne \textsc{crc}32(p_0)$ **then reject**
    return p_0

3.2 LG-CISH coding channels

The framed bytes f of Eqn (33) are turned into codebook indices by one of two lossless codes. In both cases f is first mapped to a single non-negative integer, with a `0x01` sentinel byte prepended so that leading zero bytes are preserved through the integer round-trip,

$$v = \textsc{int}_{\mathrm{BE}}(\texttt{0x01} \,\|\, f)\,. \tag{35}$$

Channel 1: base-N positional coding. The integer v is written in radix N, most-significant digit first, so that

$$v = \sum_{t=0}^{L-1} d_t\, N^t, \qquad d_t \in \{0, \dots, N-1\}, \quad L = \lfloor \log_N v \rfloor + 1, \tag{36}$$

and digit d_t selects image c_{d_t}. Because Eqn (36) is a bijection, decoding simply reads the digits back; this channel attains the ceiling of Eqn (16), at the cost of allowing an image to repeat. As a small illustration, suppose $N = 40$ and the integer to be sent is $v = 1\,000\,000$. Repeated division by 40 gives the digits $(9, 12, 20, 0)$, since $1\,000\,000 = 9 \cdot 40^3 + 12 \cdot 40^2 + 20 \cdot 40 + 0$, so the four images c_9, c_{12}, c_{20}, c_0 are emitted. The receiver, having recovered those indices, re-forms $9 \cdot 40^3 + 12 \cdot 40^2 + 20 \cdot 40 + 0 = 1\,000\,000$ and thereby recovers v exactly.

Channel 2: Lehmer-permutation coding. To obtain a cover in which no image repeats within a block, the integer v is instead written in radix $m = P(N, B)$ (Eqn (23)), giving block values $(e_0, \dots, e_{L'-1})$. Each block value is expanded into B distinct images by a mixed-radix (Lehmer) code, in which the t-th image of a block is identified by its rank r_t among the still-available images (radix $N - t$):

$$u = \sum_{t=0}^{B-1} r_t \prod_{j=t+1}^{B-1} (N - j), \qquad 0 \le r_t < N - t. \tag{37}$$

A naïve Lehmer code maps a small leading block value to an almost-sorted image prefix, which is a visible artefact. It is removed by *whitening* each block value with a deterministic, shared keystream before expansion,

$$\tilde{e}_i = (e_i + \mu_i) \bmod m, \qquad \mu_i = \textsc{sha}256(\text{“perm-whiten-}i\text{”}) \bmod m, \tag{38}$$

which is invertible modulo m and preserves both losslessness and within-block distinctness. The two directions of the block code are given in Algorithm 3.

Algorithm 3 Lehmer block coding (permEncode / permDecode)

function permEncode(u, N, B) ▷ *integer → B distinct indices*
  for $t = B-1$ **down to** 0: $r_t \leftarrow u \bmod (N-t)$; $u \leftarrow \lfloor u/(N-t) \rfloor$
  $A \leftarrow [0, 1, \dots, N-1]$; **return** $[\,A.\mathrm{pop}(r_t)$ for $t = 0 \dots B-1\,]$
function permDecode$(\langle a_0, \dots, a_{B-1} \rangle, N)$ ▷ *indices → integer*
  $A \leftarrow [0, \dots, N-1]$; $u \leftarrow 0$
  for $t = 0$ **to** $B-1$: $r \leftarrow A.\mathrm{index}(a_t)$; $u \leftarrow u(N-t) + r$; $A.\mathrm{pop}(r)$
  return u ▷ *Eqn* (37)

The mechanics of Eqn (37) are best seen on a small example. Consider a codebook of $N = 4$ images and a block of $B = 4$, for which $P(4, 4) = 4! = 24$ orderings are available and a block therefore carries $\log_2 24 \approx 4.585$ bits. Suppose the whitened block value is $\tilde{e} = 17$. The ranks are read out by successive division with decreasing radix: $17 = 2 \cdot (3!) + 5$, then $5 = 2 \cdot (2!) + 1$, then $1 = 1 \cdot (1!) + 0$, giving the rank vector $(r_0, r_1, r_2, r_3) = (2, 2, 1, 0)$. Consuming these ranks from the available list $A = [0, 1, 2, 3]$ yields, in order, the indices $2, 3, 1, 0$; that is, the block emits images c_2, c_3, c_1, c_0, which are manifestly distinct. Decoding inverts the process: the rank of each emitted index among the remaining images is recomputed, and the mixed-radix sum of Eqn (37) reconstructs $\tilde{e} = 17$, from which the whitening of Eqn (38) is removed. The example makes concrete the two guarantees of the code — exact invertibility and within-block distinctness.

The choice between the two channels is a deployment decision rather than a correctness one, since both are lossless. Base-N coding is preferred when the overriding objective is to minimise the number of transmitted images, because it attains the ceiling of Eqn (16); permutation coding is preferred when the cover must survive human inspection as a plausible album, because the no-repeat guarantee removes the conspicuous artefact of the same image appearing several times in a short sequence. The block size B interpolates smoothly between the two regimes: a small block sacrifices little capacity relative to Eqn (16) while still forbidding repeats within the block, whereas the full-block case $B = N$ maximises the no-repeat window at the largest capacity cost. In this way Eqn (24) exposes a single, continuous control that a deployment can set according to whether compactness or plausibility is the more pressing requirement.

3.2.1 LG-CISH encoding

The complete sender procedure is assembled in Algorithm 4. The message is framed by Eqn (32)–33, mapped to the integer v by Eqn (35), converted to indices by either Eqn (36) or the whitened Lehmer code of Eqn (38), and finally realised as images. When the optional alias mechanism of §3.3 is enabled, the image chosen for a given index is the alias that best improves cover plausibility; since every alias of an index decodes to the same index, this choice does not affect recovery.

For the decoder to invert the procedure, a small set of protocol parameters must be agreed in advance: whether compression and encryption were applied, which coding channel was used, and, for permutation coding, the block size B. These parameters are not secret — by Assumption 1.1 the warden may know them — and they occupy a negligible number of bits beside the payload; in a fixed deployment they are simply constants shared by both parties. With them fixed, Algorithm 4 is fully determined and its inverse, Algorithm 5, is unambiguous.

Algorithm 4 ENCODE: message $\rightarrow$ ordered image sequence

1: **Input:** message m; codebook $\mathcal{C}$ (size N); key κ; flags *perm*, block B, *alias*
2: $f \leftarrow \textsc{wrap}(m, \kappa)$ ▷ *Algorithm 2*
3: $v \leftarrow \textsc{int}_{\mathrm{BE}}(\texttt{0x01} \,\|\, f)$ ▷ *Eqn* (35)
4: **if** *perm* **then**
5: $\quad m' \leftarrow P(N, B)$; $(e_0, \dots) \leftarrow \textsc{toBaseN}(v, m')$
6: $\quad \mathrm{idx} \leftarrow \textsc{concat}_i\ \textsc{permEncode}\big((e_i + \mu_i) \bmod m', N, B\big)$ ▷ *Eqn* (38)
7: **else** $\mathrm{idx} \leftarrow \textsc{toBaseN}(v, N)$ ▷ *Eqn* (36)
8: **if** *alias* **then** $S \leftarrow [\,\textsc{chooseAlias}(j)$ for $j \in \mathrm{idx}\,]$ **else** $S \leftarrow [\,c_j$ for $j \in \mathrm{idx}\,]$
9: **return** image sequence S

3.2.2 Lossless decoding

The receiver reverses the encoder exactly. Each received image is embedded with CLIP and matched to the nearest codebook entry by Eqn (17), which recovers the transmitted index whenever the decoding margin of Eqn (18) is positive; under Definition 3.1 and adequate transport stability this holds for every position, so the index stream is recovered without error. The recovered indices are then passed back through the inverse coding (Eqn (36) or the un-whitened Lehmer inverse of Eqn (37)–38) to reconstruct the integer v, the sentinel byte is stripped to obtain f, and UNWRAP verifies the CRC and inverts the encryption and compression. The procedure is given in Algorithm 5.

Proposition 3.1 (End-to-end losslessness)**.** If every index is recovered correctly, i.e. $\gamma_t > 0$ for all t in Eqn (18), then $\textsc{Decode}(\textsc{Encode}(m)) = m$ exactly and BER $= 0$. This follows because each stage — Eqn (32), Eqn (33), Eqn (35), Eqn (36), and Eqn (37)–38 — is a bijection and is inverted in reverse order.

Algorithm 5 DECODE: image sequence → message

```
Input: received images ⟨ỹ_0, ..., ỹ_{T−1}⟩; codebook C; key κ; flags perm, B
for t = 0 to T − 1:  î_t ← arg max_j s(e(ỹ_t), C_j)                          ▷ Eqn (17)
if perm then
    m′ ← P(N, B);  for each block i:  e_i ← (permDecode(î_[iB:iB+B], N) − μ_i) mod m′
    v ← fromBaseN((e_0, ...), m′)
else  v ← fromBaseN(î, N)
f ← stripSentinel(bytes_BE(v))                                              ▷ inverse of Eqn (35)
return unwrap(f, κ)                                          ▷ verify CRC, decrypt, decompress
```

Computational cost. The asymmetry between the two directions of the scheme is worth noting. Encoding is dominated by big-integer arithmetic: the base-N conversion of Eqn (36) costs $\mathcal{O}(L)$ big-integer operations for a sequence of L images, and the cryptographic frame is linear in the payload size; no neural network is evaluated on the sender side, so encoding is effectively instantaneous. Decoding is dominated by the feature map: each of the L received images is passed once through the CLIP encoder and then compared against the N codebook embeddings, giving a cost of

$$\mathcal{O}\big(L \cdot \big(C_\Phi + Nd\big)\big)\,, \tag{39}$$

where C_Φ is the cost of a single CLIP forward pass and $d = 512$ is the embedding dimension. The nearest-neighbour term Nd is negligible beside the forward pass C_Φ, so decoding time scales essentially linearly in the number of received images and is governed by the throughput of the image encoder. The memory footprint is a single $N \times d$ matrix of codebook embeddings, which is a few megabytes for the codebooks considered here and is independent of message length.

Behaviour beyond the operating range. It is instructive to consider what happens when a channel degradation exceeds the range that the codebook was provisioned to absorb. In that regime the margin of Eqn (18) can turn negative at one or more positions, and by Eqn (19) the message-level success probability falls sharply. Crucially, the scheme does not fail silently: because the recovered indices no longer invert to a byte string whose trailing CRC is consistent, the integrity check of §3.1.2 rejects the message. The failure is thereby made *visible*, and the standard recourse — retransmission, or a request for a less aggressive channel — is available. This behaviour is the practical realisation of the principle argued in §1.5, that a detected error is far preferable to an undetected one, and it is the reason the CRC is retained even when encryption is disabled.

3.3 Plausibility aliases

The undetectability argument of §1.6 concerns the *marginal* distribution of each transmitted image, which is naturally innocent. A separate, weaker concern is whether the *sequence* of images looks like a plausible personal album to a human observer. To address this without touching losslessness, each codebook slot j is optionally equipped with a small set of visually interchangeable candidate images, called aliases, that have been verified by a vision-capable large language model to depict the same subject as c_j. All aliases of slot j map to the same index j, so the decoder is entirely unaffected. During encoding, whenever index j is to be emitted, the alias that best improves the coherence of the running sequence is selected. Because the choice among aliases is invisible to the index-recovery rule of Eqn (17), cover plausibility is improved as a free, orthogonal degree of freedom, and Proposition 3.1 continues to hold.

In more detail, the alias set for each slot is prepared offline by a generate-and-verify loop. Candidate images that plausibly depict the same subject as c_j are proposed — either retrieved from the database or synthesised — and each candidate is then submitted, together with the anchor image c_j, to a vision-capable large language model that is asked to confirm that the two depict the same subject and to score their interchangeability. Only candidates whose confirmation passes a fixed threshold are retained, so that the alias relation is conservative by construction. Because this verification is performed once, offline, it adds nothing to the per-message encoding cost; and because the resulting aliases all share the slot index j, they remain invisible to the recovery rule of Eqn (17). It is stressed that aliases affect only the human plausibility of the *ordering* of the cover, not the distributional undetectability of each individual image, which is already guaranteed by Eqn (14).

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